

Clustering Countries on Sustainable Energy

A New Era of Energy



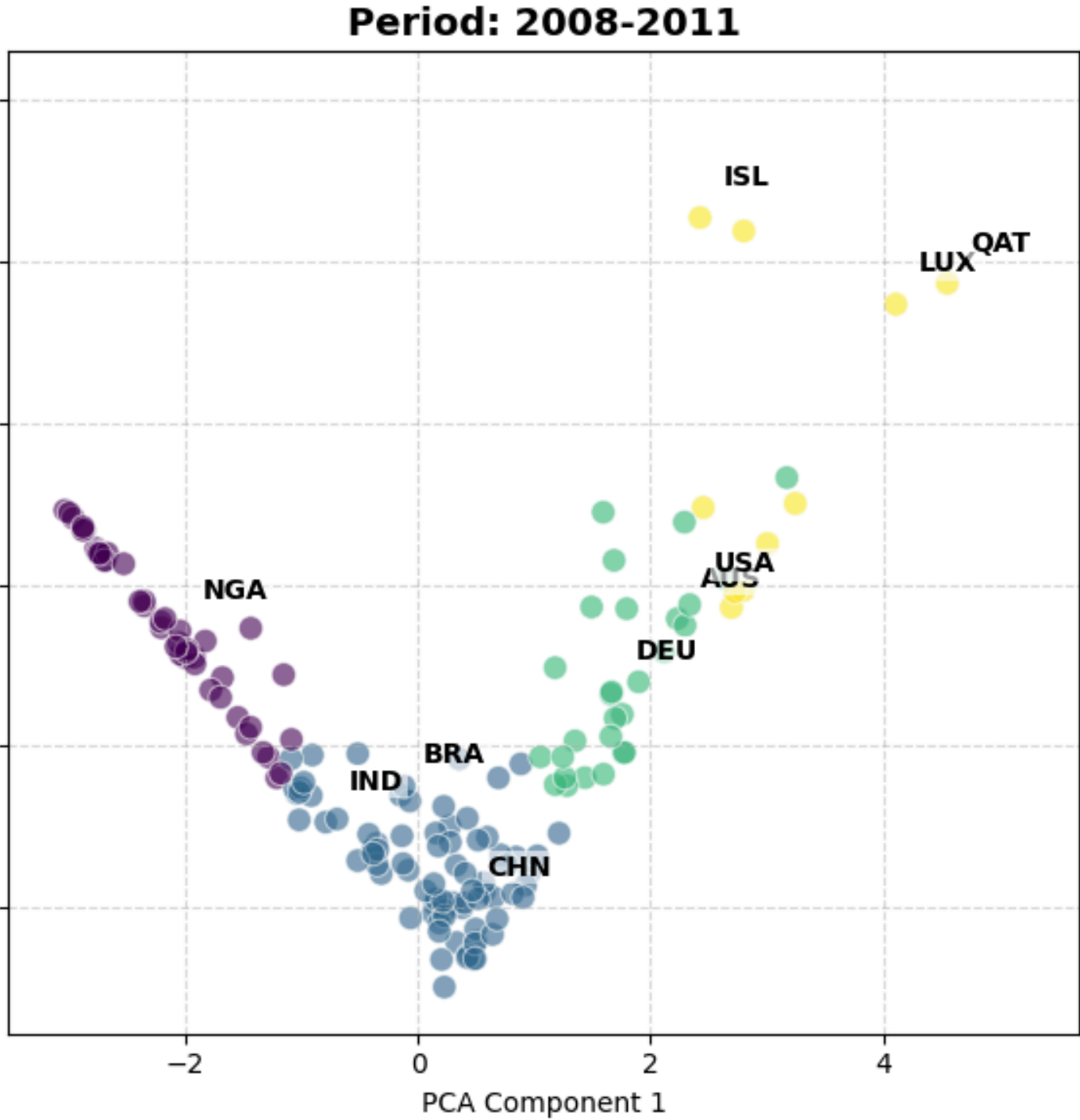
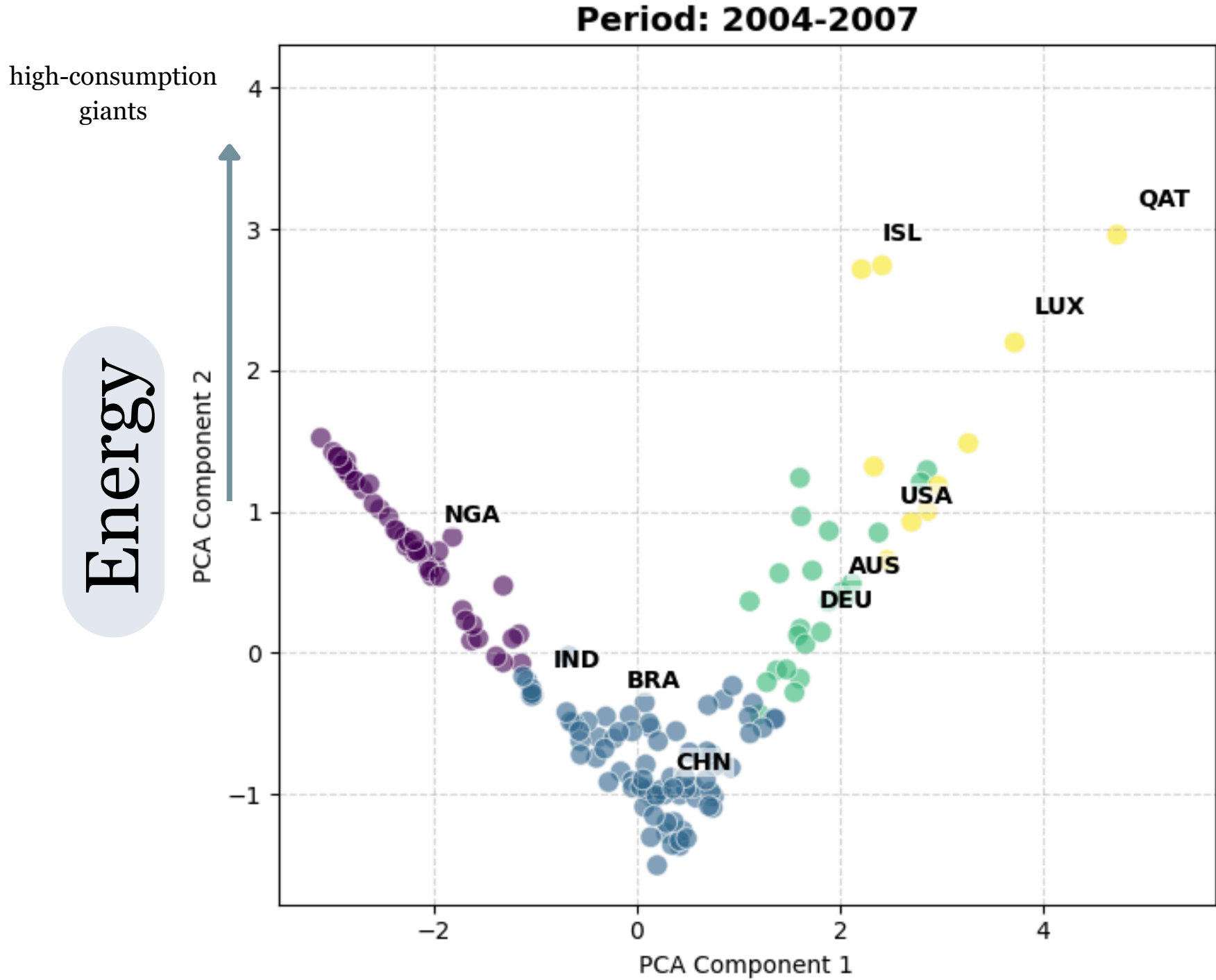
Objectives of the Project

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Mapping the World's Energy Archetypes

- Classify countries by their actual energy behavior.
- The Method: We applied K-Means Clustering on 4 key metrics (GDP, Renewable Share, Consumption, Access) across 16 years.
- The Result: 4 distinct global profiles emerged:
 1. **Energy Deficient:** Struggling with basic access (e.g., Sub-Saharan Africa).
 2. **Emerging Transitioners:** Rapidly industrializing engines (e.g., China, Brazil).
 3. **Fossil Giants:** Wealthy, high-consumption, fossil-reliant (e.g., USA, Australia).
 4. **Green Leaders:** Wealthy but efficient (e.g., Germany, UK).

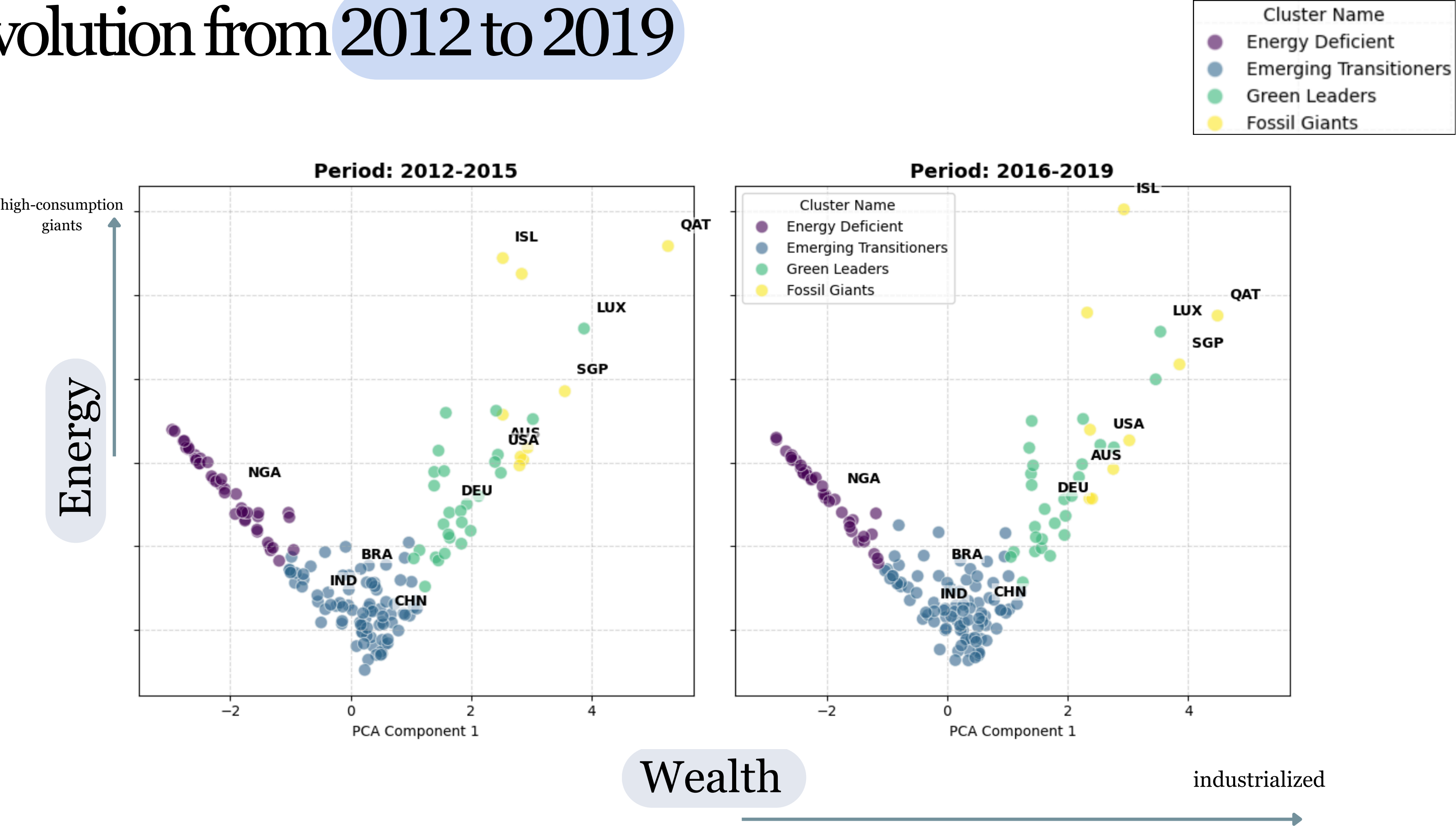
Evolution from 2004 to 2011



Wealth

industrialized

Evolution from 2012 to 2019



The "Price vs. Efficiency" Impact



Summary

- Data Preparation & Structuring
- Exploration of Trends & Distributions
- Analysis of Relationships & Correlations
- Synthesis & Answering the Key Question

Key Descriptive Statistics (Quick Overview)

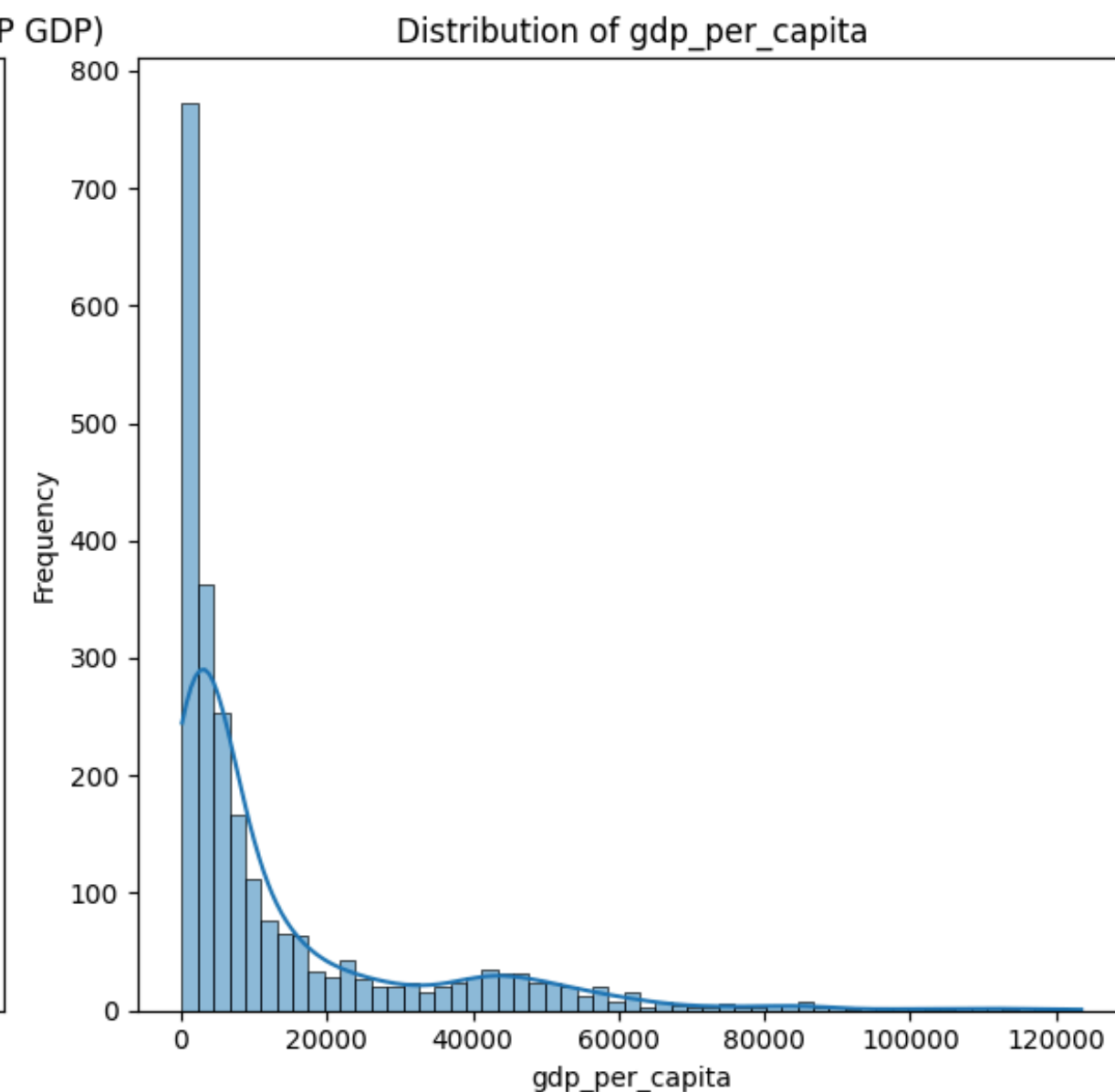
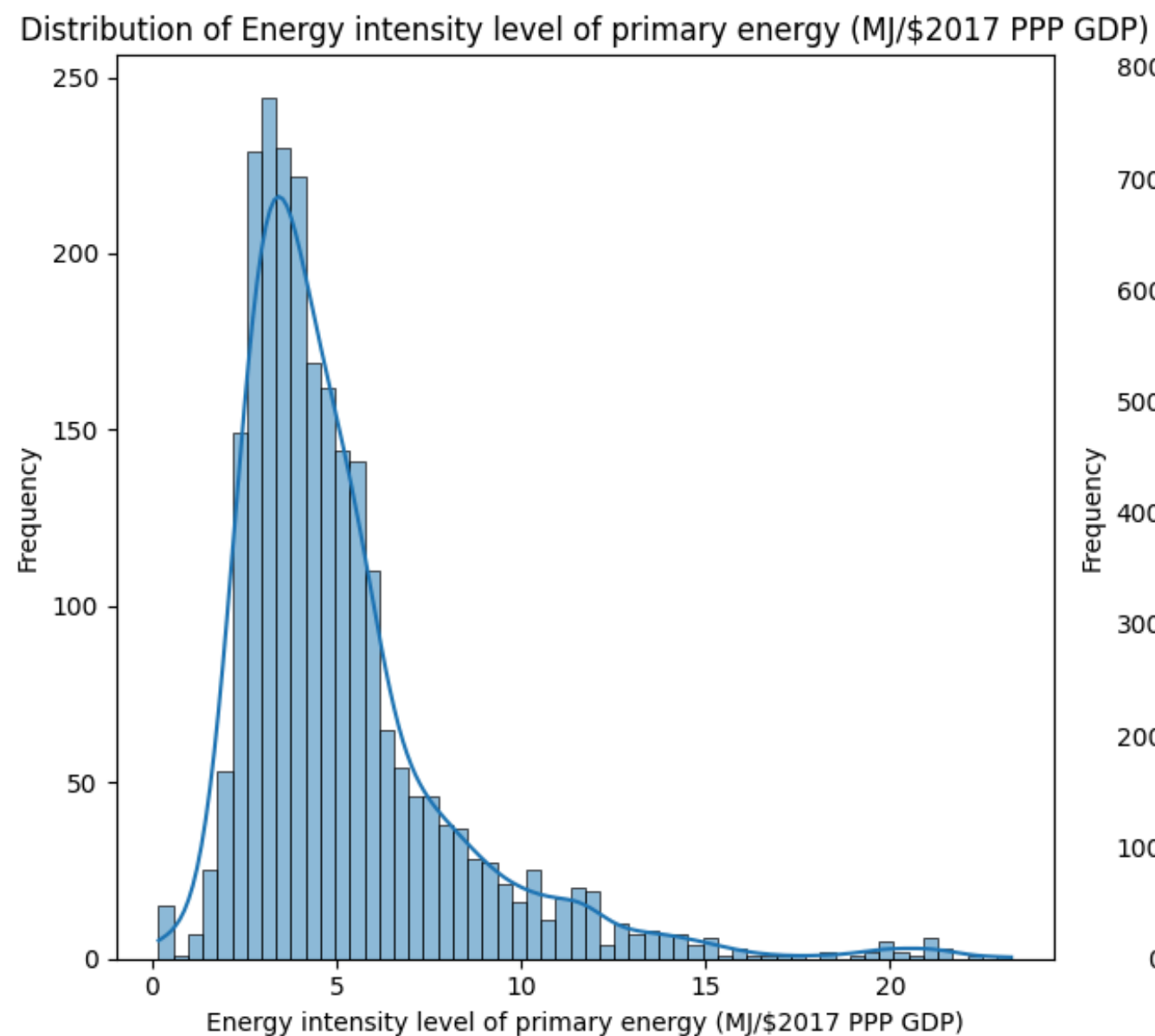
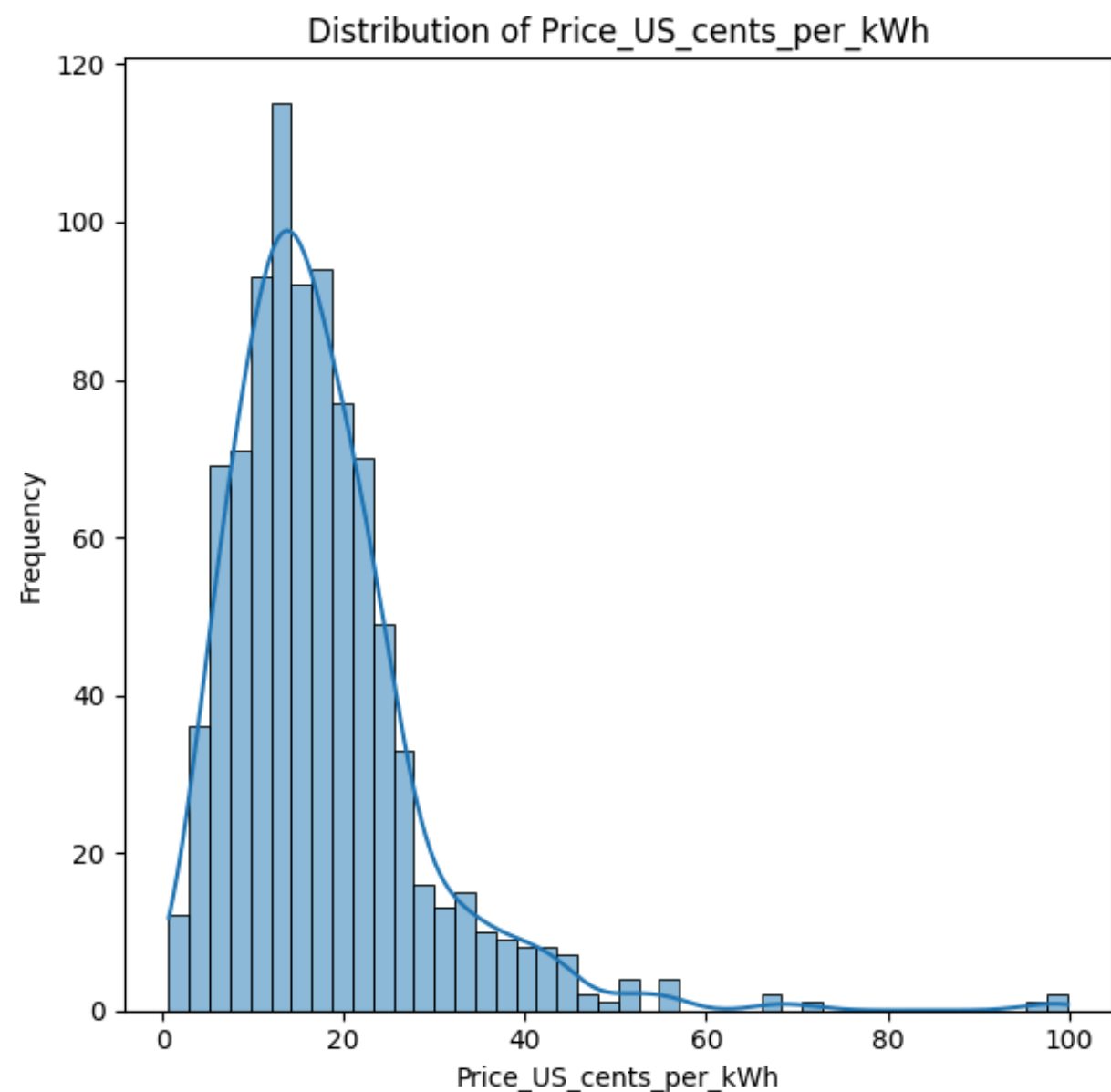
Data Preparation & Structuring

Data Preparation

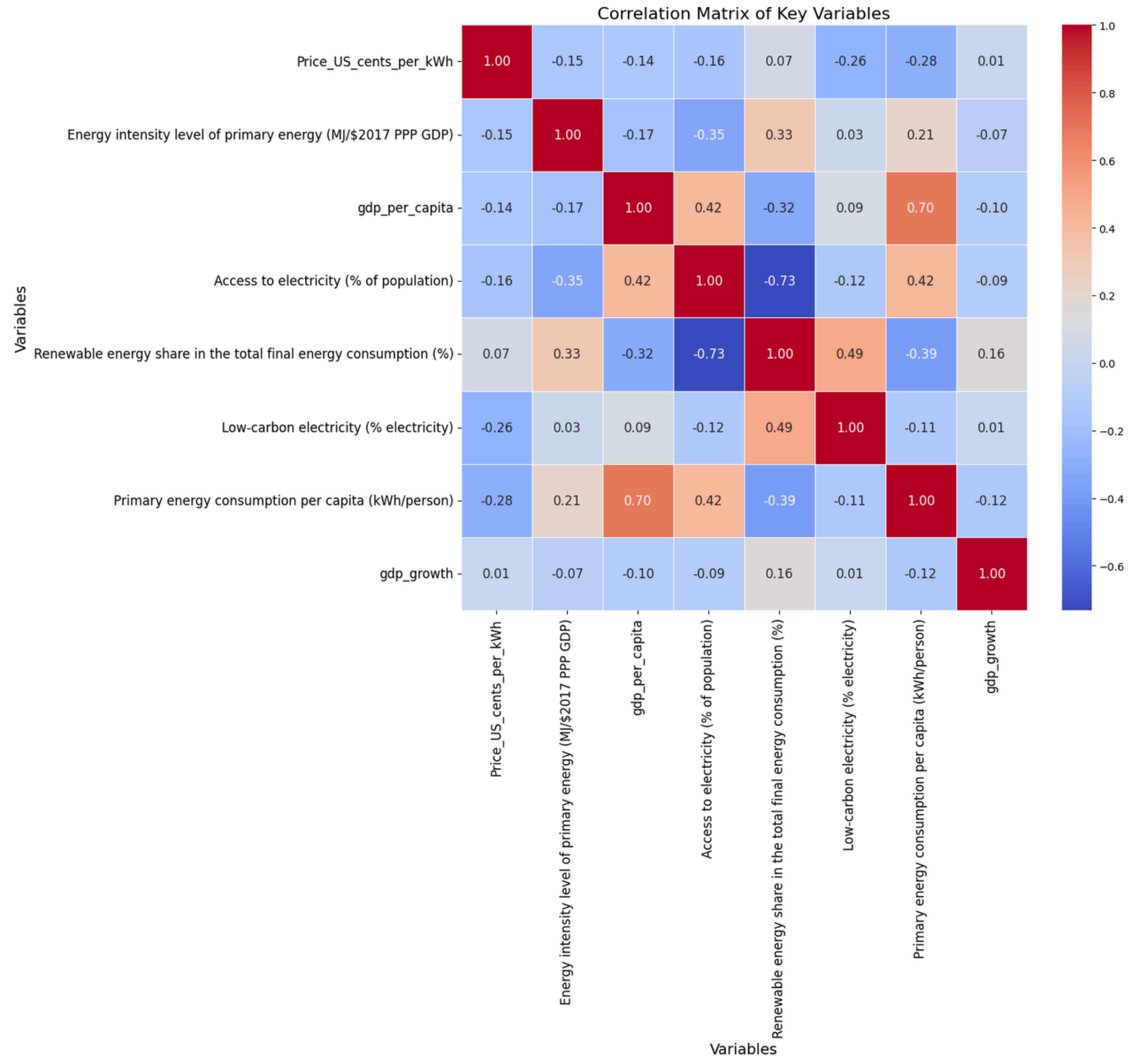
- Data Origin
- Strategic Merging
- Essential Data Cleansing
- Objective

Key Descriptive Statistics (Quick Overview)			
Statistic	Price_US_cents_per_kWh	Energy intensity level of primary energy (MJ/\$2017 PPP GDP)	gdp_per_capita
Count	914	2456	2443
Mean	17.56	5.13	13689.10
Median (50%)	15.60	4.31	5109.85
Minimum (Min)	0.70	0.18	128.34
Maximum (Max)	99.90	23.29	123514.00

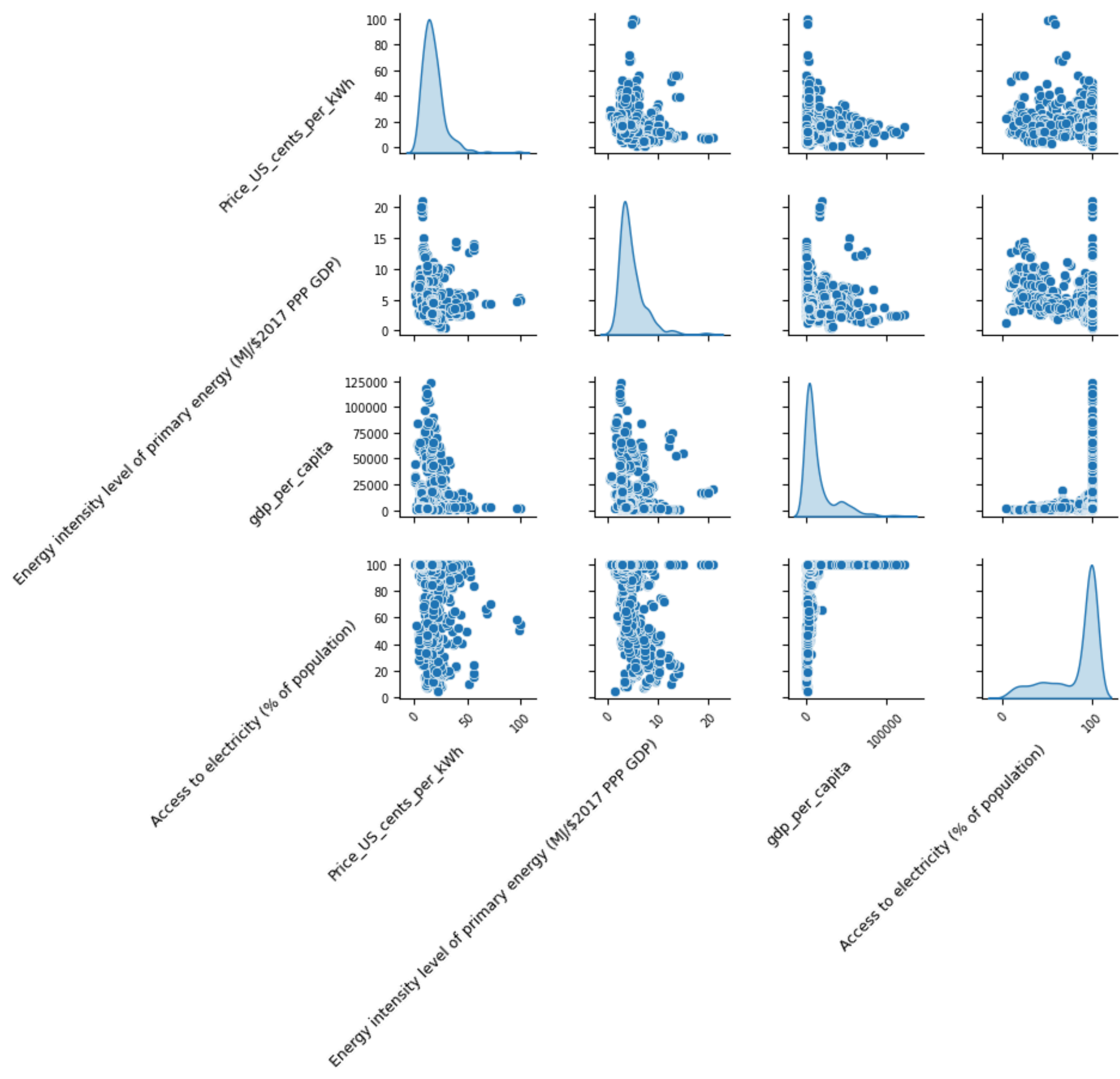
Exploration of Trends & Distributions



Analysis of Relationships & Correlations

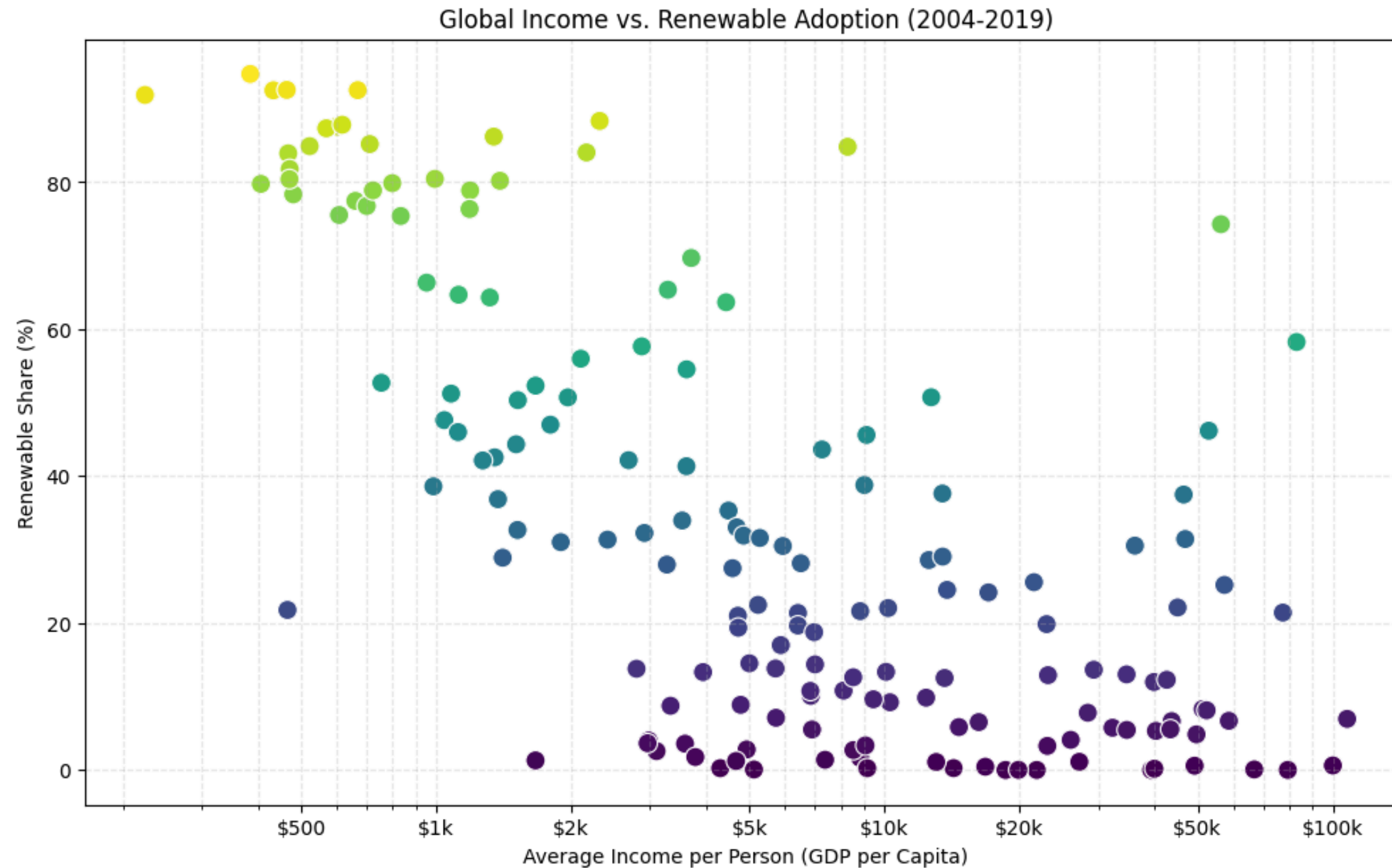


Pair plot diagram

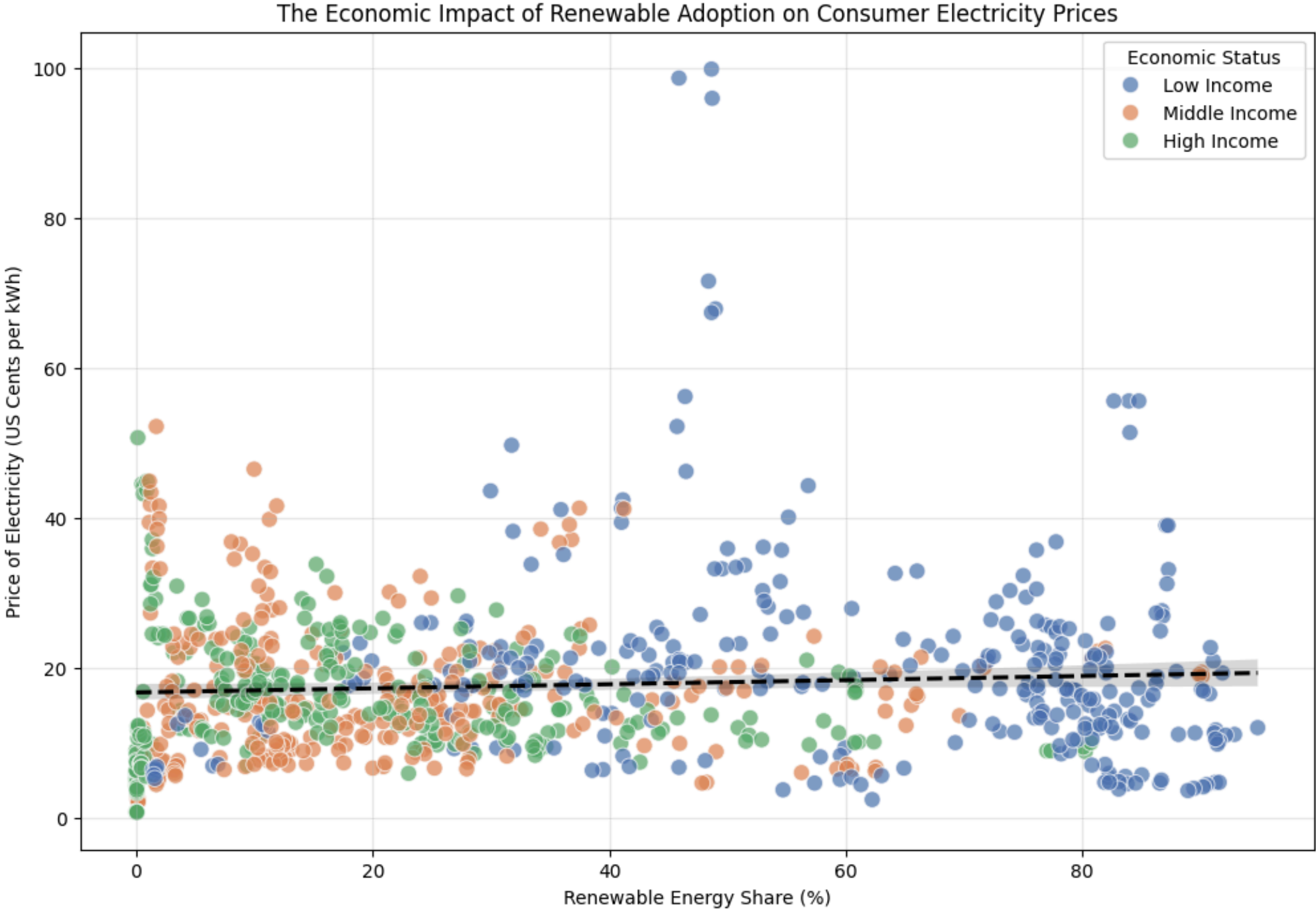


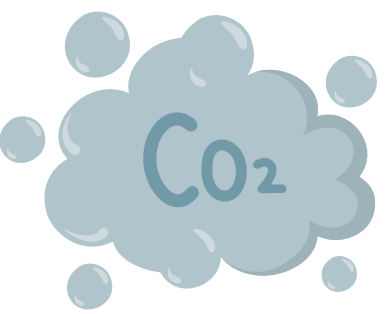
Analysis of the Relationship between Growth and Renewable Energy Adoption

Economic



Price Sensitivity Analysis: Renewable Share vs. Consumer Cost



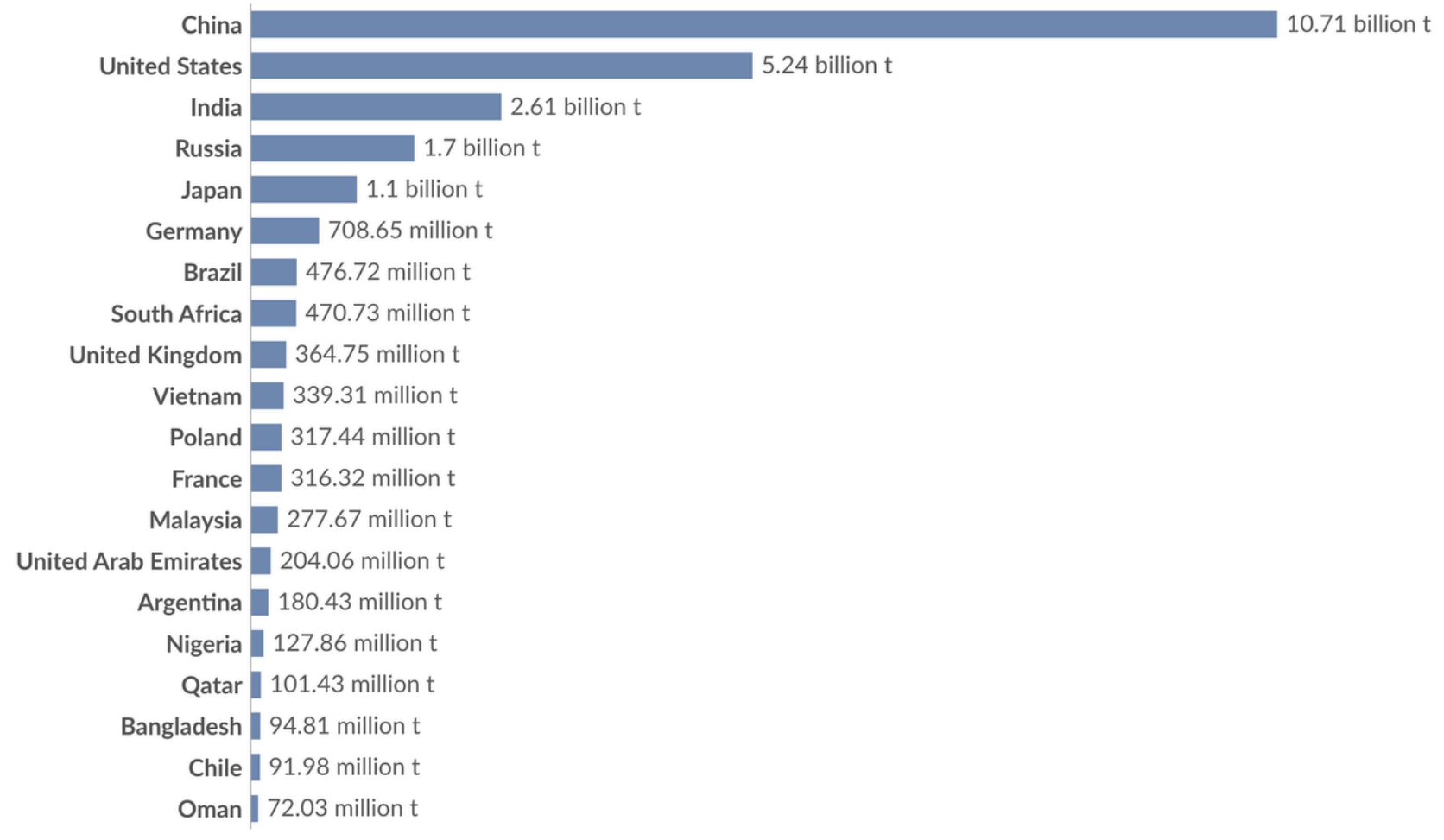
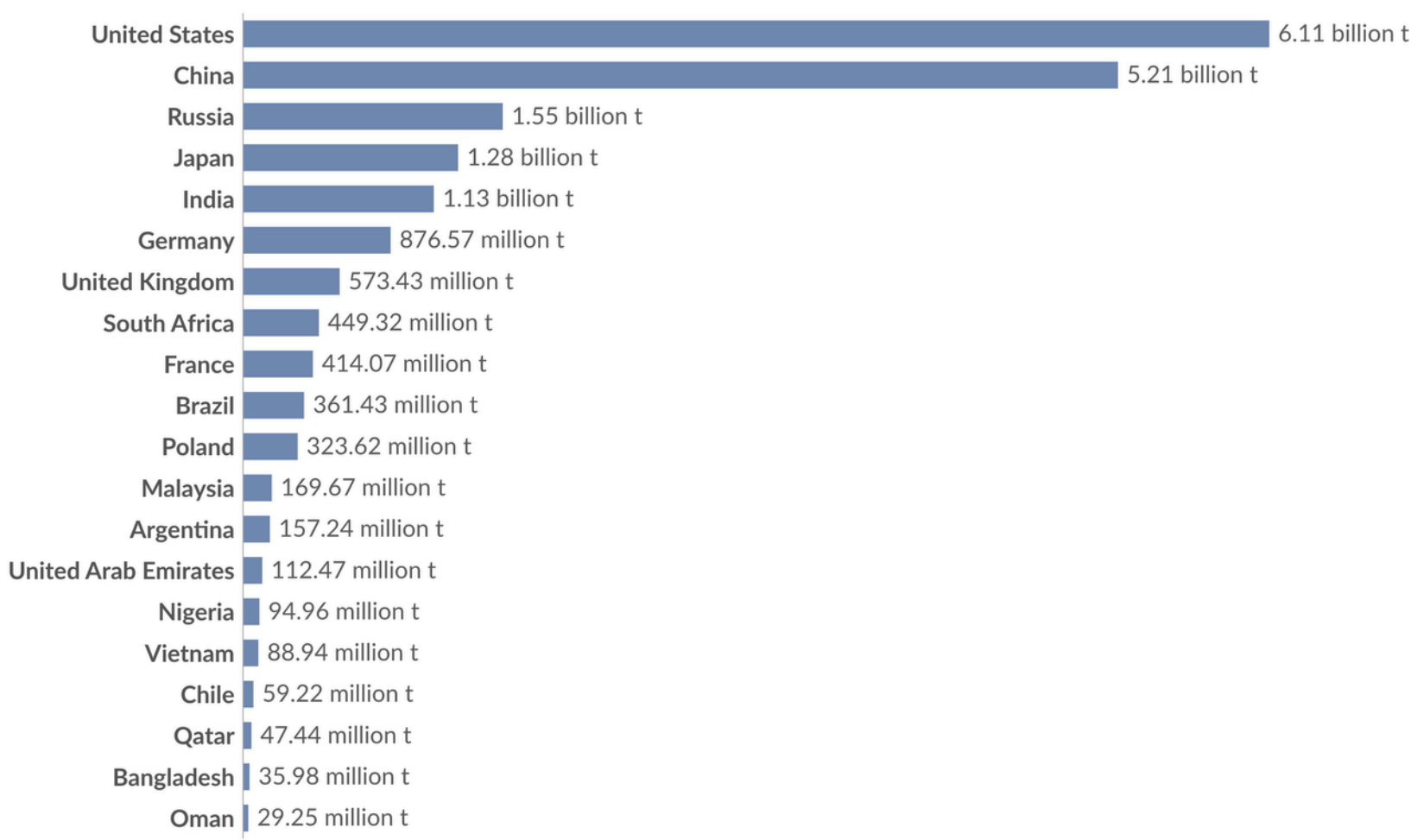


Annual CO2 emissions

2004

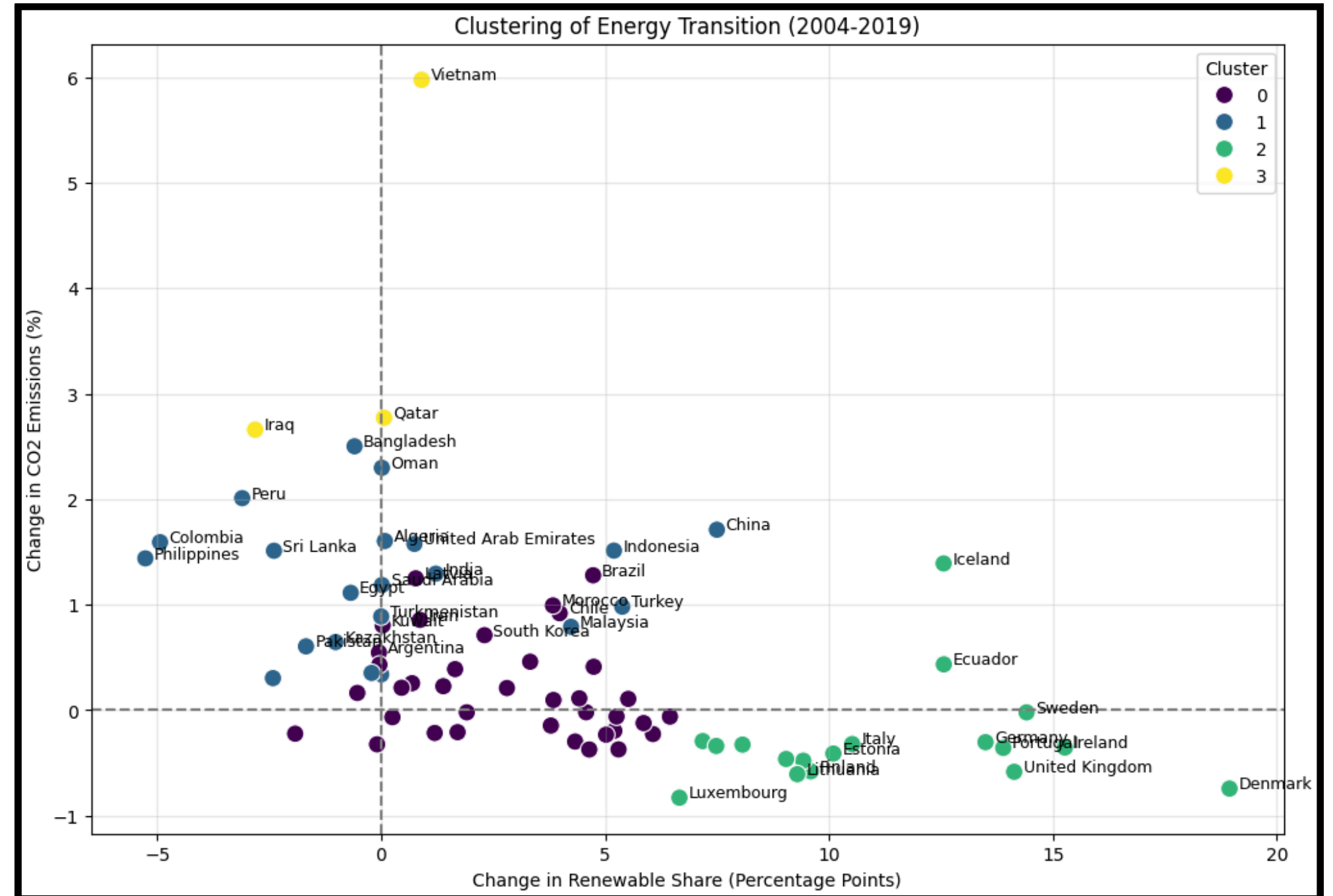


2019



Explication of the link between energy

- Link between renewable energy and change in CO2
- Link with based economy factor



Conclusion



- Objective 1 - “The map”
- Objective 2 - “The economic driver”
- Objective 3 - “The policy check”
- Objective 4 - “The trend”

Correlation :

- Price & efficiency
- Aid & Green transition
- Wealth & Speed





Thank you for your attention!